

Project Overview

Project Objective

To develop a **secure biometric-based verification system** that enables:

- Student identity verification at exam centers
 - Integration with biometric devices (iris, fingerprint, face)
 - Aadhaar-based authentication workflow
 - Centralized data management and monitoring
 - Real-time verification logs and reporting
 - Multi-platform access (Web + Mobile Application)
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System Architecture Overview

The system will consist of:

- Web application for administration and monitoring
 - Mobile/tablet application for on-ground verification
 - Biometric device integration layer
 - Backend API services
 - Centralized database
 - Aadhaar authentication integration (as per compliance)
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Project Modules & Scope

1. Authentication & Role Management

- Secure login system (JWT / token-based)
 - Roles:
 - Super Admin
 - Center Admin
 - Operator (Tablet User)
 - Role-based permissions and access control
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2. Student Data Management

- Bulk student data import (Excel/CSV)
 - Fields:
 - Name
 - Aadhaar number (masked/secure)
 - Exam details
 - Biometric references (if pre-registered)
 - Data validation and deduplication
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3. Exam & Center Management

- Create and manage exams
 - Assign students to exam centers
 - Map devices and operators to centers
 - Scheduling and session control
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4. Biometric Device Integration

- Integration with registered biometric devices:
 - Fingerprint scanners
 - Iris scanners
 - Camera (for face authentication)
 - SDK/API integration with hardware providers
 - Device registration with system
 - Device health and connectivity tracking
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5. Mobile / Tablet Application

- Login for center operators
 - Student verification interface
 - Biometric capture:
 - Fingerprint
 - Iris
 - Face
 - Real-time authentication response
 - Offline/low-network handling (sync mechanism)
 - Capture logs and timestamps
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6. Aadhaar Verification Integration

- Secure Aadhaar authentication workflow
 - API integration (UIDAI-compliant if applicable)
 - OTP / biometric-based verification (as per regulations)
 - Data encryption and compliance handling
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7. Verification Workflow

Operational flow:

1. Operator logs into tablet
 2. Selects exam and center
 3. Searches student / scans ID
 4. Captures biometric (fingerprint / iris / face)
 5. System verifies:
 - Local database match
 - Aadhaar authentication (if enabled)
 6. Displays result:
 - Verified / Not Verified
 7. Stores log with timestamp and device ID
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8. Admin Dashboard (Web Application)

- Centralized monitoring panel

- View:
 - Student verification status
 - Center-wise activity
 - Device status
 - Real-time logs and analytics
 - Search and filter records
 - Export reports
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9. Logs & Audit System

- Complete audit trail:
 - Login logs
 - Verification logs
 - Device activity
 - Timestamped records
 - Fraud detection indicators (multiple attempts, mismatch)
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10. Notification & Alerts

- Alerts for:
 - Device offline
 - Failed verifications
 - Suspicious activity
 - Notification channels:
 - Email
 - SMS / WhatsApp (optional integration)
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11. Security & Compliance

- Data encryption (at rest and in transit)
 - Aadhaar data masking and compliance
 - Secure API communication
 - Role-based data restriction
 - Logging for audit and compliance
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12. Backend API Development

- Scalable REST APIs
 - Integration with biometric SDKs
 - Authentication and authorization
 - Data processing and validation
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13. Deployment & Infrastructure

- Cloud deployment (AWS / DigitalOcean)
 - Secure server setup
 - Load handling for concurrent verifications
 - Backup and recovery setup
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Timeline Breakdown (1.5 Months)

Week 1

- Requirement analysis
 - System architecture design
 - Database schema design
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Week 2

- Backend development (core APIs)
 - Authentication and user management
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Week 3

- Student data import system
- Exam and center management
- Initial mobile app setup

Week 4

- Biometric device integration
 - Mobile app verification workflow
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Week 5

- Aadhaar integration
 - Admin dashboard development
 - Logs and reporting system
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Week 6

- Testing (functional and performance)
 - Security implementation
 - Bug fixing
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Final 0.5 Month

- Deployment
 - Final validation
 - Project delivery
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Web Application Development

Mobile Application Development

Biometric Device Integration

Aadhaar Integration & Compliance

Testing & Quality Assurance

Deployment & DevOps

Project Management

Technology Stack

Web Application:

- React.js

Mobile Application:

- React Native / Flutter

Backend:

- Node.js (Express.js)

Database:

- MongoDB / PostgreSQL

Biometric Integration:

- Device SDKs (vendor-specific)

Security:

- JWT Authentication
- SSL Encryption

Cloud:

- AWS / DigitalOcean
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Assumptions

- Biometric devices provide SDK/API access
 - Aadhaar integration follows legal compliance requirements
 - Internet connectivity available at exam centers
 - Device drivers compatible with tablet/mobile OS
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Exclusions

- Hardware procurement (biometric devices)
- Government certification or licensing costs
- Third-party API charges (Aadhaar, SMS, WhatsApp)
- On-site deployment support